Verbatim recall is pervasive and stable across individuals

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Memory for linguistic input serves as the basis for online prediction and dependency formation, making it a key component of language processing. Such memory is thought to be essentially compressive [1]. However, humans can spontaneously recall verbatim sequences at impressive delays [2], suggesting both compressed and verbatim memories of linguistic experience are in principle available. Practically successful LLM architectures make use of both abstract, higher-order embeddings of past context (akin to compressed encodings), as well as token-by-token copying mechanisms (akin to verbatim encodings) [3], suggesting both types of representation are useful for language prediction. But is direct verbatim recall prevalent enough to be functionally relevant in language, as it appears to be in LLMs? Here we test this by mining a large-scale story re-telling dataset [4]. Our results suggest *yes*: We find that incidental verbatim recall is pervasive in spontaneous story retelling. We further observe non-trivial interindividual systematicity in what sequences are remembered, indicating that such verbatim learning is related to intrinsic properties of a linguistic stimulus [5-6].

We analyzed the *Naturalistic Free Recall Dataset* [4-5], which consists of 4 spoken stories, ranging from 1-2k words each, and the recalls of 229 participants. To create this dataset, participants were asked to listen to two stories and, after each one, to 'recollect the story in as much detail as possible by speaking aloud.' The resulting recollections, approximately 327k words in total, were transcribed and normalized for case and punctuation. To find verbatim spans, we used Greedy String Tiling [7] to first identify overlapping trigram anchor sequences in the source and its recall. Then, the preceding and following 20 tokens around the trigram tiles were extracted. We applied the Smith-Waterman algorithm [8] to find the optimal local sequence alignment between these (an algorithm which generalizes Levenshtein distance for local sequence alignment, finding maximal, aligned sequences and allowing for some mismatches and gaps). Thus we identified both strict verbatim and 'semi-verbatim' sequences (i.e. with some gaps or mismatches).

We found that participants' recall contained much verbatim material, even without explicit instruction [cf. 9]. Approximately 10% of words recalled by participants belonged to verbatim or semi-verbatim spans of five words or more, similar to the rate [2] observed with instructions to 'read [a] story back .. accurately' on much shorter stories. Figure 1 shows the distribution, by participant, indicating that most stories were effective at eliciting verbatim recall. Median length of recall was 7.3 words (semi-verbatim) & 5.7 words (verbatim) (see Fig. 2).

Verbatim productions were also bursty and strongly correlated across participants. We analyzed their temporal structure by projecting recall strength onto fixed-width word bins. Within each bin, recall strength was defined as the maximum overlap with any recalled span, giving a sparse, participant-specific recall trace. Heatmaps visualize these traces (Fig. 3). Intersubject correlations (ISC; via leave-one-out cross-validation), revealed reliable recall structure shared across participants within each story (mean r, 0.28-0.48, $p \sim 0$, via permutation; Fig. 4). This suggests that the structure of the story itself was critical in driving verbatim "hotspots."

Finally, we investigated whether unexpected text was more likely to elicit verbatim recall, using surprisals extracted from a Pythia-family transformer model [11]. Using the same binned approach as above, we found a reliable *negative relationship* between surprisal (aligned & lagged) and the likelihood of verbatim recall (log-odds: [-0.12_{Lag0}, -0.10_{Lag1}, -0.09_{Lag2}]; ps < .05, via permutation), as well as a reduction in cross-participant synchronization. This corroborates experimental work on recall [11], where foiled expectations or low-likelihood continuations can disrupt encoding, but contrasts with the effect of surprisal on recognition memory [12]. However, surprisal may travel with coherence in this naturalistic dataset, frustrating firm conclusions about underlying cause at this stage. On-going work is thus investigating other predictors, including event segmentation, narrative structure, and valence, as well as lexical and prosodic factors.

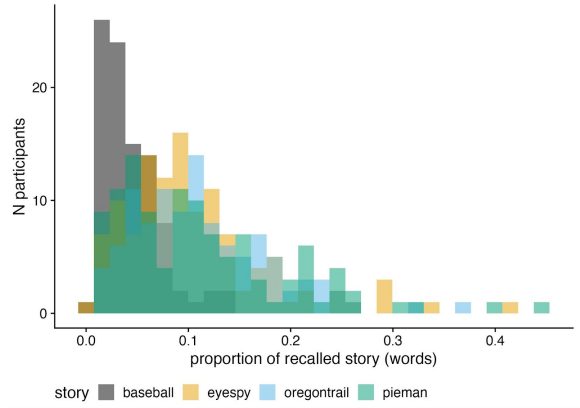

Figure 1 Distribution of verbatim recalled spans (greater than 5 words), as proportion of story length.

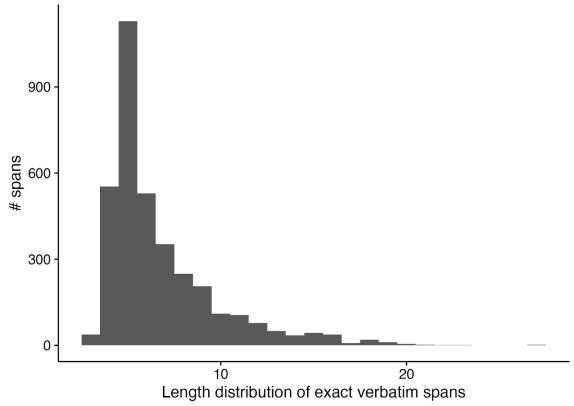

Figure 2 Length distribution of (exact) verbatim recalled spans.

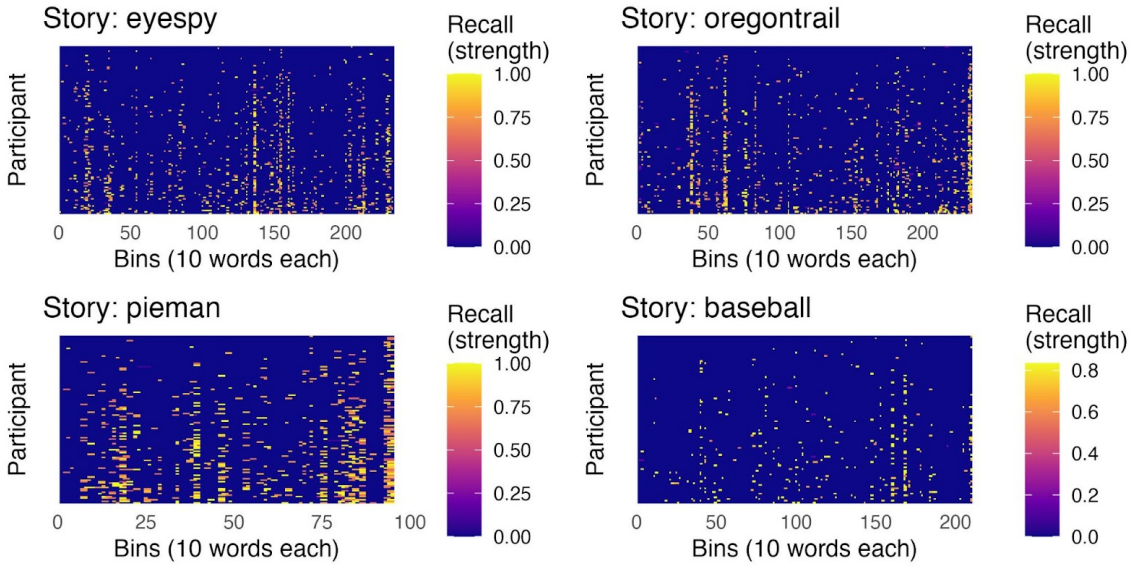

Figure 3 Participant-level recall traces (rows) across story position (columns). Color indicates the maximum recall strength within fixed-width word bins.

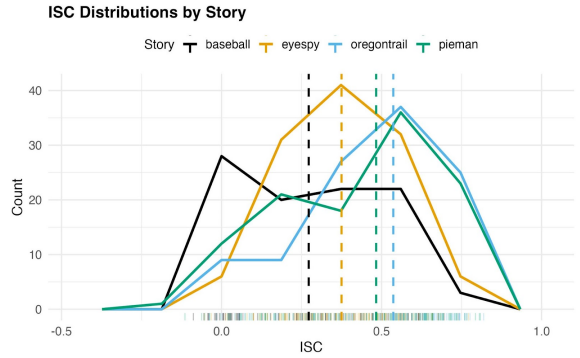

Figure 4 Intersubject correlations (ISC) by story. Dashed lines mark median ISCs and x-axis rug ticks the individual scores.

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